



Chapter 10 FIR Digital Filter Design

- *Preliminary Consideration*
- *FIR Digital Filter Design Methods*
- *MATLAB Functions in FIR Filter Design*



10.1 Preliminary Consideration



- **Digital Filter Specifications**
- **Basic Approach to FIR Digital Filter Design**
- **FIR Digital Filter Order Estimation**

10.1.1 Basic Approaches to design FIR filters



- Unlike IIR digital filter design, FIR filter design does not have any connection with the design of analog filters.
- The design of FIR filters is based on a direct approximation of the specified magnitude response, with often added requirement that the phase response be linear.



- Recall a causal FIR frequency response $H(e^{j\omega})$ of length $N+1$:

$$H(e^{j\omega}) = \sum_{n=0}^N h[n] e^{-j\omega n} = |H(e^{j\omega})| e^{j\theta(\omega)}$$

Magnitude Response

Phase Response

- For a linear-phase FIR, we have:

$$\theta(\omega) = c\omega + \beta, \quad \text{For } c = -\frac{N}{2}, \beta = 0, \pi, \pm\pi/2$$

- Or, as we known: $h[n] = \pm h[N-n]$

10.1.2 Estimation of the Filter Order



- The order (length-1) of $G(z)$ is directly estimated from the digital filter specification.
- Several authors have advanced formulas :
- **Kaiser's Formula:**

$$N \cong \frac{-20 \log_{10}(\sqrt{\delta_P \delta_S}) - 13}{14.6(\omega_S - \omega_P) / (2\pi)}$$

- **Bellanger's Formula:**

$$N \cong -\frac{2}{3} \frac{\log_{10}(10\delta_P \delta_S)}{(\omega_S - \omega_P) / (2\pi)} - 1$$

Estimation of the Filter Order



- **Hermann's Formula:**

$$N \cong \frac{D_{\infty}(\delta_p, \delta_s) - F(\delta_p, \delta_s)[(\omega_s - \omega_p) / 2\pi]^2}{(\omega_s - \omega_p) / 2\pi}$$

$$D_{\infty}(\delta_p, \delta_s) = \left[a1(\log_{10} \delta_p)^2 + a2(\log_{10} \delta_p) + a3 \right] \log_{10} \delta_s \\ - \left[a4(\log_{10} \delta_p)^2 + a5(\log_{10} \delta_p) + a6 \right],$$

$$F(\delta_p, \delta_s) = b1 + b2 \left[\log_{10} \delta_p - \log_{10} \delta_s \right]$$

**a1=0.005309, a2=0.07114, a3=-0.4761, a4=0.00266,
a5=0.5941, a6=0.4278, b1=11.01217, b2=0.51244**



- **A Comparison of FIR Filter Order Formulas:**

- (1) Each of these three formulas provide only **an estimate** of the required filter order.
- (2) The estimated filter order N of the FIR filter is **inversely proportional to the transition band width** ($\omega_p - \omega_s$). And does not depend on the actual location of the transition band.
- (3) The order of Kaiser's and Bellanger's formulas depends on the product δ_p and δ_s .
- (4) Each one of three formulas can also be used to estimate the order of highpass, bandpass and bandstop FIR filter.
- (5) For two transition bands, we always use the smallest width to compute the order of the filter.

10.2 FIR Filter Design Based on Windowed Fourier Series



- Two direct approaches to design of FIR filters:
 - (1) **Truncating** the Fourier series representation of the prescribed frequency response. (direct and straightforward)
 - (2) **Sampling the frequency response** to get the DFT of $h[n]$, and use IDFT to get $h[n]$.

10.2.1 Least Integral-Squared Error Design of FIR Filters



- Let $H_d(e^{j\omega})$ denote the desired frequency response function, $h_d[n]$ denotes the impulse response samples, so :

$$H_d(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_d[n]e^{-j\omega n}$$

Fourier
series

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega, -\infty \leq n \leq \infty$$

In most applications, for meeting the FR $H_d(e^{j\omega})$ requirements, $h_d[n]$ is infinite.



We want to find: a Length-N, causal

$$h_t[n] \xrightarrow{\text{approximation}} h_d[n]$$

So:

$$H_t(e^{j\omega}) = \sum_{n=0}^{N-1} h_t[n] e^{-jn\omega} \xrightarrow{\text{approximation}} H_d(e^{j\omega})$$

The procedure of using $h_t[n]$ to replace $h_d[n]$ is called truncating or windowing.

Using what approximating criterion?

Minimum Integral-Squared Error.

- Define integral-squared error:

$$\Phi_R = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H_t(e^{j\omega}) - H_d(e^{j\omega})|^2 d\omega$$

Using Parseval's relation, we get:

$$\begin{aligned} \Phi_R &= \sum_{n=-\infty}^{\infty} |h_t[n] - h_d[n]|^2 \\ &= \sum_{n=-M}^M |h_t[n] - h_d[n]|^2 + \sum_{n=-\infty}^{-M-1} h_d^2[n] + \sum_{n=M+1}^{\infty} h_d^2[n] \end{aligned}$$

When $h_t[n] = h_d[n]$ for $-M \leq n \leq M$, the Φ_R is minimum.

⇒ Best finite-length approximation to ideal infinite-length impulse response in the mean-square sense is obtained by truncation.



- A causal FIR filter with an impulse response $h[n]$ can be derived from $h_t[n]$ by delaying:

$$h[n] = h_t[n - M]$$

The causal FIR filter $h[n]$ has the same magnitude response as $h_t[n]$ and its phase response has a linear phase shift of ωM radians with respect to that of $h_t[n]$.

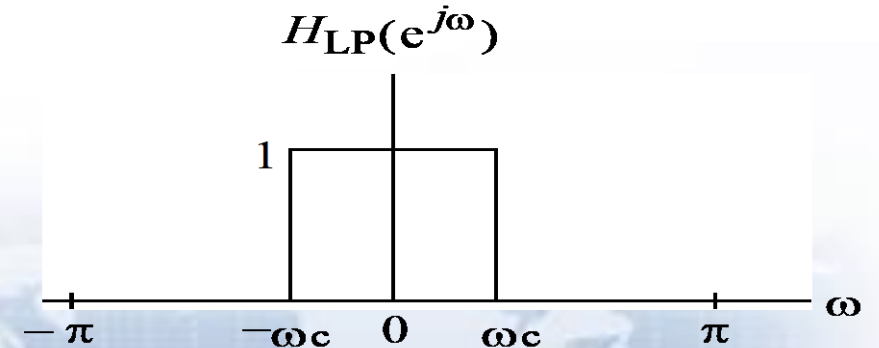
- $h[n]$ ---- linear phase
- $h_t[n]$ ----zero phase

10.2.2 Impulse Responses of Ideal Filters



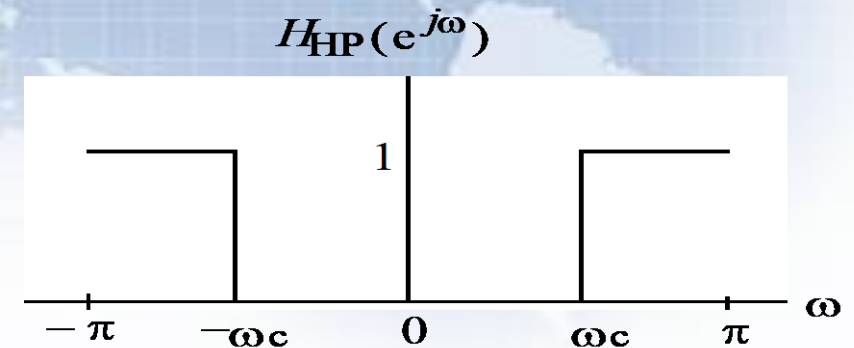
- Ideal lowpass filter:

$$h_{LP}[n] = \frac{\sin \omega_c n}{\pi n}, -\infty \leq n \leq \infty$$



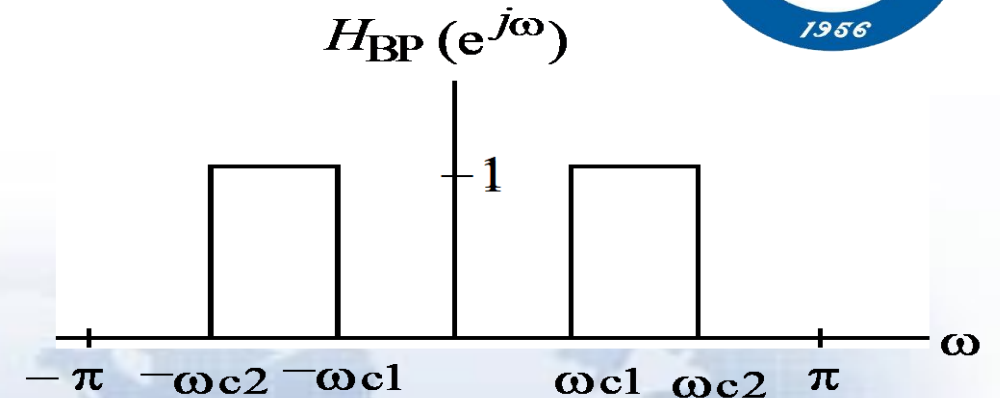
- Ideal highpass filter:

$$h_{HP}[n] = \begin{cases} 1 - \frac{\omega_c}{\pi}, & \text{for } n = 0 \\ -\frac{\sin \omega_c n}{\pi n}, & \text{for } |n| > 0 \end{cases}$$



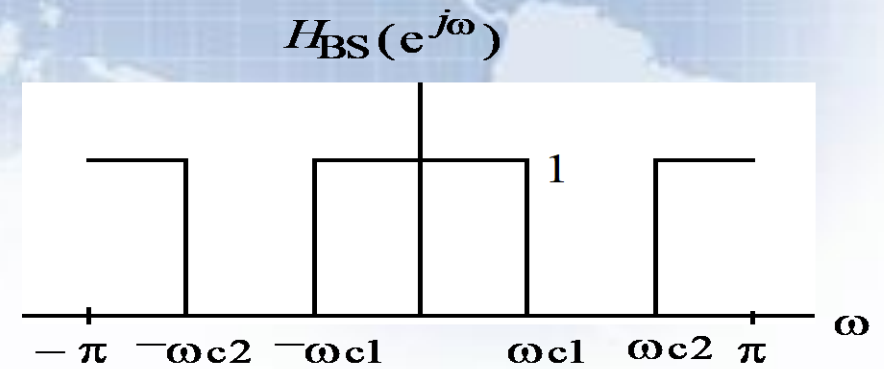
- **Ideal bandpass filter:**

$$h_{BP}[n] = \frac{\sin \omega_{c_2} n}{\pi n} - \frac{\sin \omega_{c_1} n}{\pi n}, |n| \geq 0$$



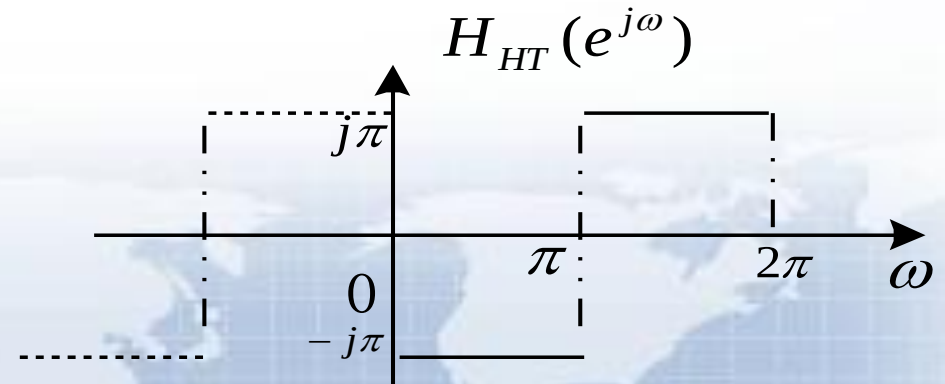
- **Ideal bandstop filter:**

$$h_{BS}[n] = \begin{cases} 1 - \frac{(\omega_{c_2} - \omega_{c_1})}{\pi}, & \text{for } n = 0 \\ \frac{\sin(\omega_{c_1} n)}{\pi n} - \frac{\sin(\omega_{c_2} n)}{\pi n}, & \text{for } |n| > 0 \end{cases}$$



- Hilbert transformer (90-degree phase shifter)

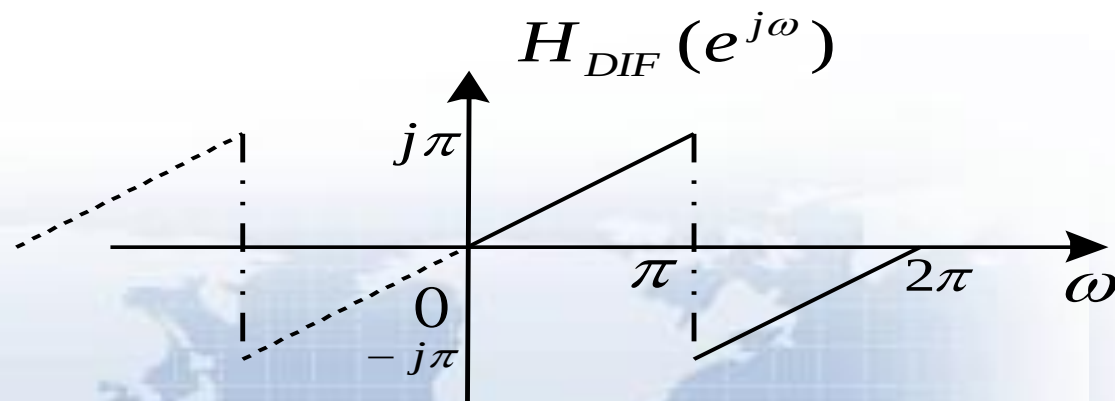
$$H_{HT}(e^{j\omega}) = \begin{cases} -j, & 0 < \omega < \pi \\ j, & -\pi < \omega < 0 \end{cases}$$



$$h_{HT}[n] = \frac{2\sin^2(\frac{n\pi}{2})}{n\pi} = \begin{cases} 0, & n = 0 \\ \frac{1}{n\pi} [1 - (-1)^n], & n \neq 0 \end{cases}$$

- Differentiator:

$$H_{DIF}(e^{j\omega}) = j\omega, 0 \leq |\omega| \leq \pi$$



$$h_{DIF}[n] = \begin{cases} 0, n = 0 \\ \frac{\cos(n\pi)}{n}, |n| \neq 0 \end{cases}$$

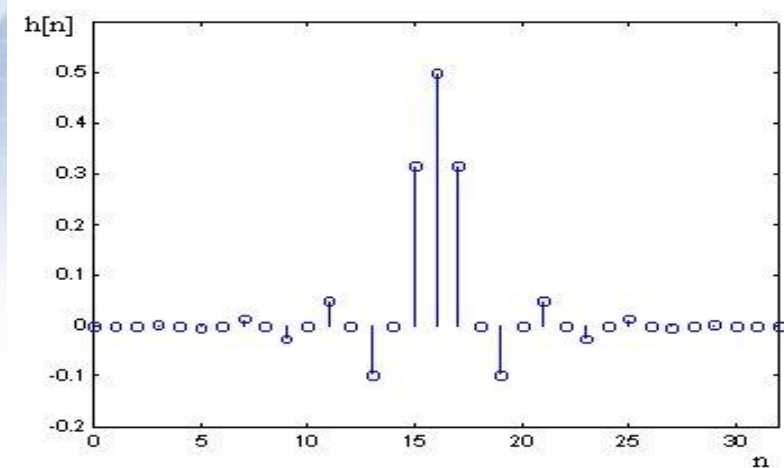
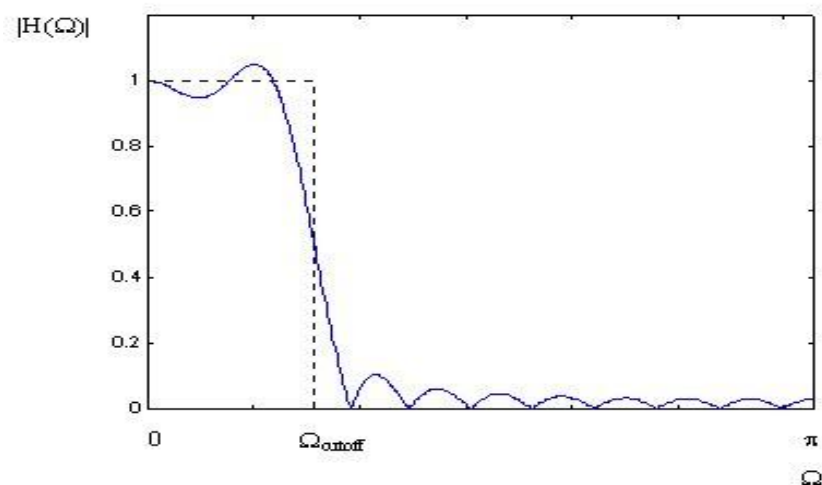
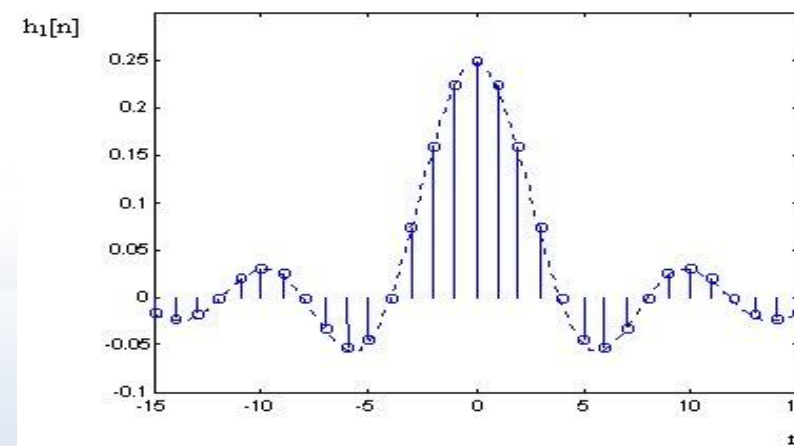
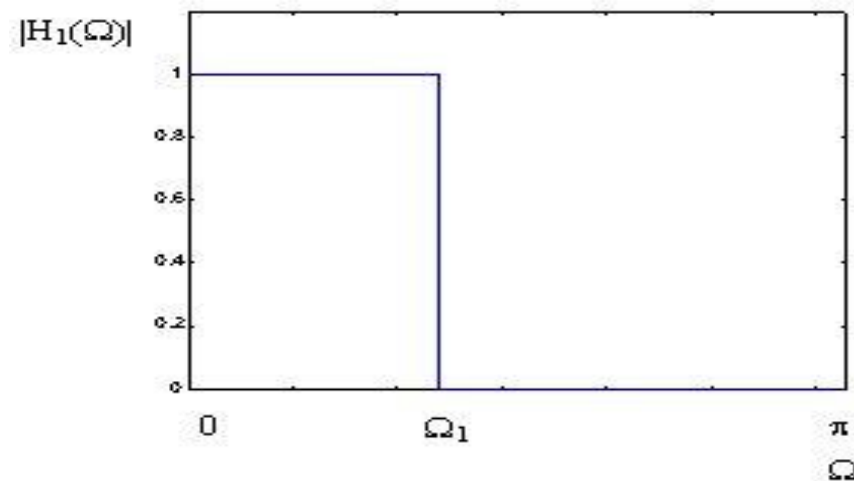
Truncate the $h[n]$ of Ideal Lowpass Filter



- The impulse response of ideal filter is infinite. By setting all coefficients outside the range $-M \leq n \leq M$ equal to zero, we arrive at a finite-length sequence.
- And by shifting to the right yields the causal FIR lowpass filter which we need.

$$\hat{h}_{LP}[n] = \frac{\sin(\omega_c(n-M))}{\pi(n-M)}, 0 \leq n \leq N-1$$

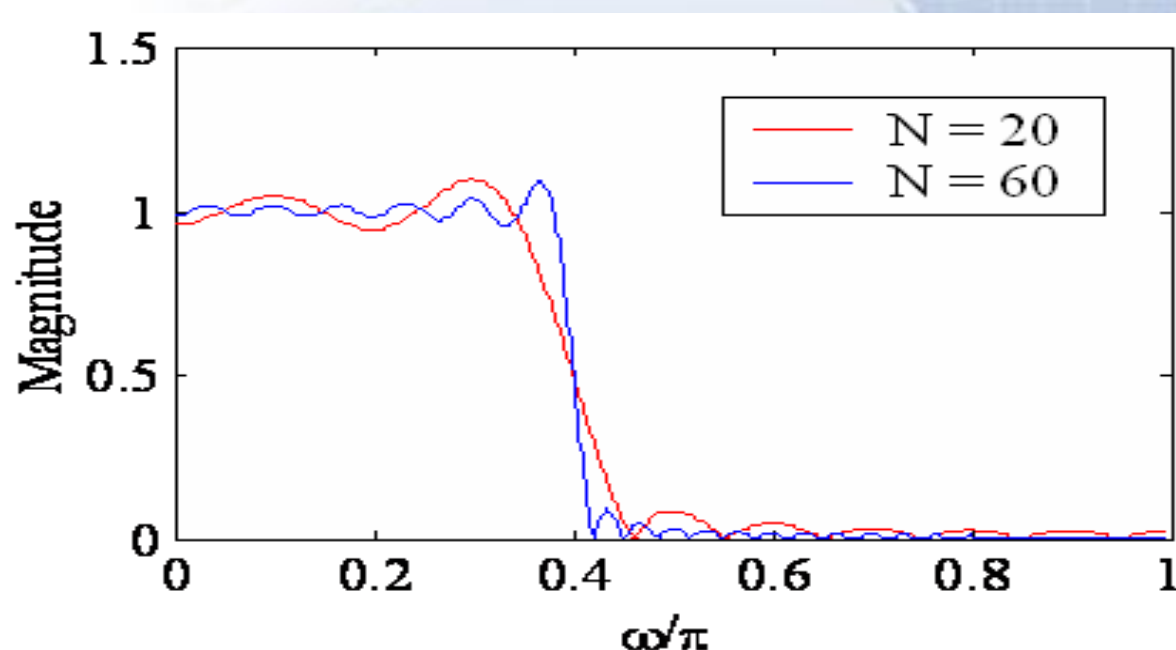
- **Note:** In above expression, length N is $2M+1$.



10.2.3 Gibbs Phenomenon -- The Effect of Truncating the $h[n]$



- Gibbs phenomenon - Oscillatory behavior in the magnitude responses of causal FIR filters obtained by truncating the impulse response coefficients of ideal filter





- As can be seen, as the length of the lowpass filter is increased, the number of ripples in both passband and stopband increases, with a corresponding decrease in the ripple widths;
- Height of the largest ripples remain the same independent of length;
→ Gibbs phenomenon
- Area under each lobe remains constant while width of each lobe decreases with an increase in M .
- Similar oscillatory behavior observed in the magnitude responses of the truncated versions of other types of ideal filters.

Gibbs Phenomenon Interpretation:



truncating operation $\rightarrow$ windowing operation:

$$h_t[n] = h_d[n] \cdot w[n]$$

In the frequency domain:

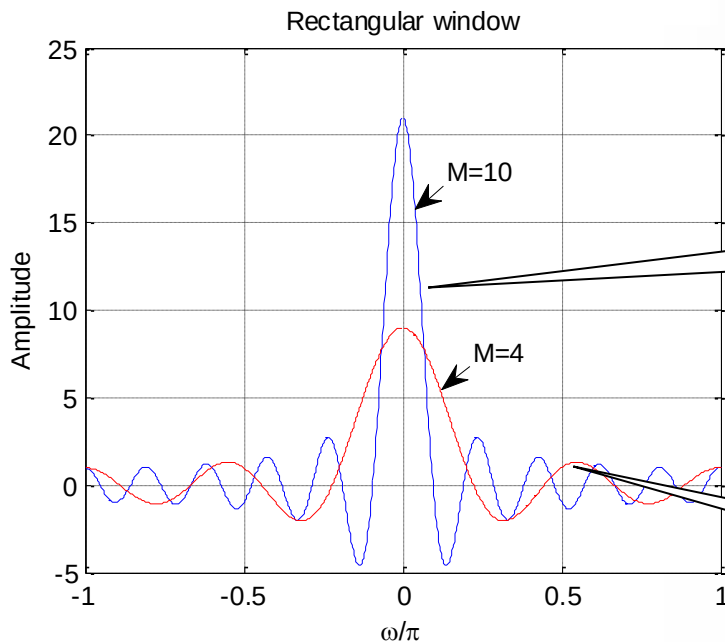
$$H_t(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\phi}) \Psi(e^{j(\omega-\phi)}) d\phi$$

where $H_t(e^{j\omega})$ and $\Psi(e^{j\omega})$ are the DTFTs of $h_t[n]$ and $w[n]$, respectively.

A rectangular window is used to achieve simple truncation:

$$w_R[n] = \begin{cases} 1, & 0 \leq |n| \leq M \\ 0, & \text{otherwise} \end{cases}$$

$$\psi_R(e^{j\omega}) = \frac{\sin(\omega(2M+1)/2)}{\sin(\omega/2)}$$

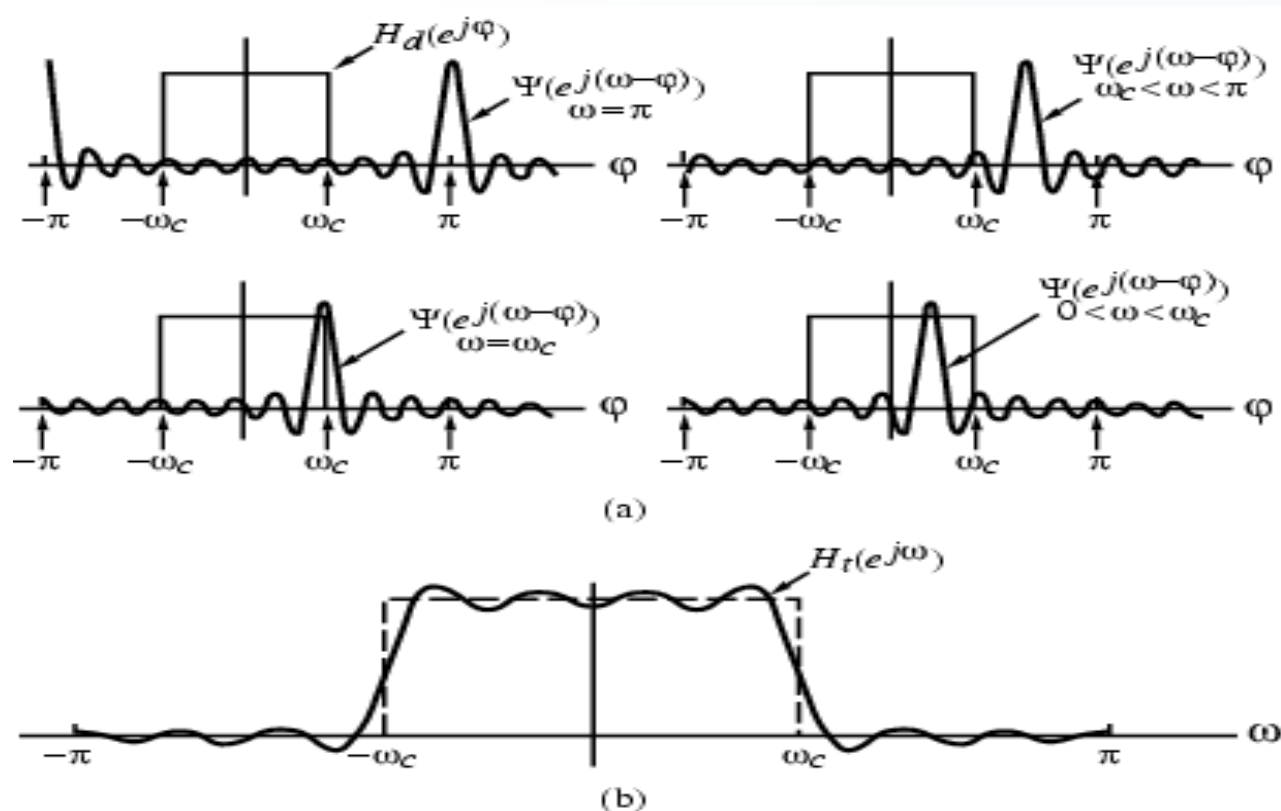


main lobe

Width: $4\pi/(2M+1)$

side-lobes

- Thus $H_t(e^{j\omega})$ is obtained by a periodic continuous convolution of $H_d(e^{j\omega})$ with $\Psi(e^{j\omega})$





- Increasing the truncation-length of window can decrease the width of the transition band, but can't change the relative value of the main-lobe and side-lobes.
- So, we need a window:
 - (1) It's main lobe width is as narrow as possible to ensure a fast transition from passband to stopband.
 - (2) Reducing the relative magnitude of the maximum sidelobe to reduce the passband and stopband ripple.



- Rectangular window has an abrupt transition to zero outside the range $-M \leq n \leq M$, which results in Gibbs phenomenon in $H_t(e^{j\omega})$
- Gibbs phenomenon can be reduced either:
 - (1) Using a window that tapers smoothly to zero at each end;
 - (2) Providing a smooth transition from passband to stopband in the magnitude specifications.

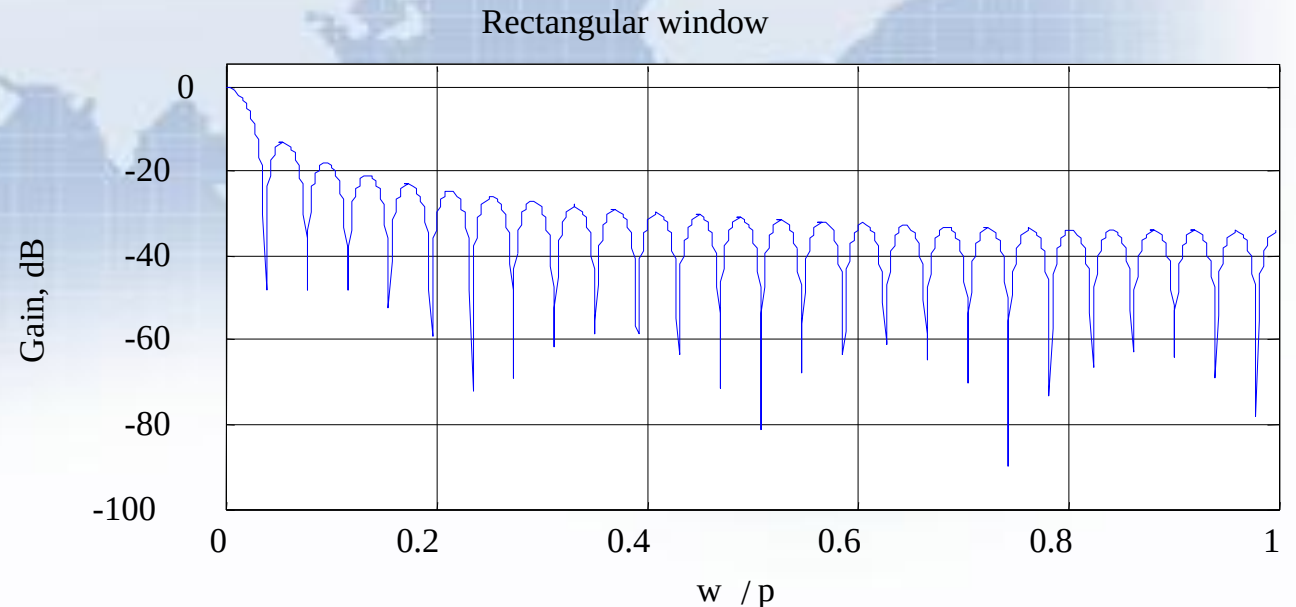
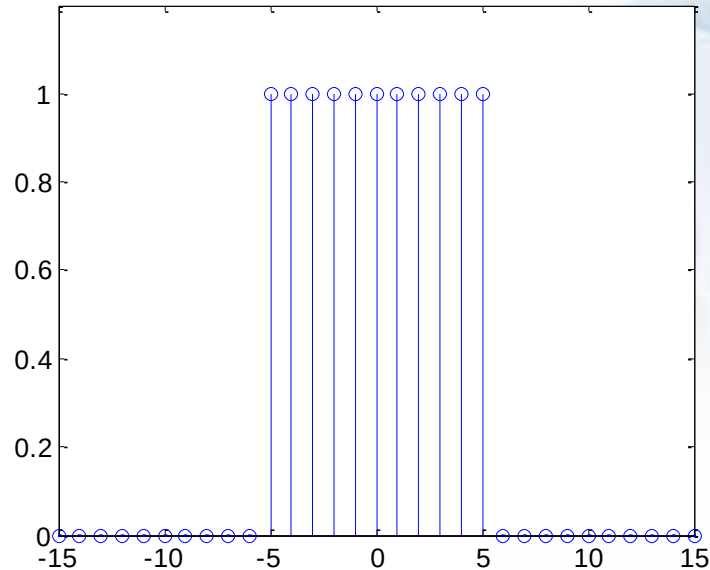
10.2.4 Fixed Window Functions



- Rectangular window

$$w_R[n] = \begin{cases} 1, & -M \leq n \leq M \\ 0, & |n| > M \end{cases}$$

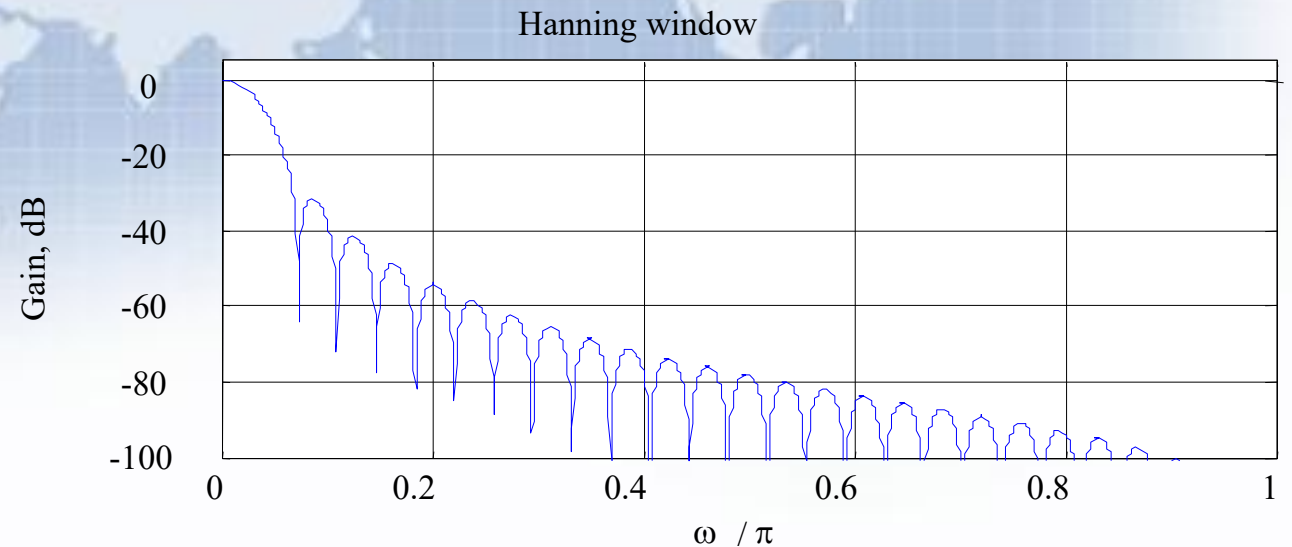
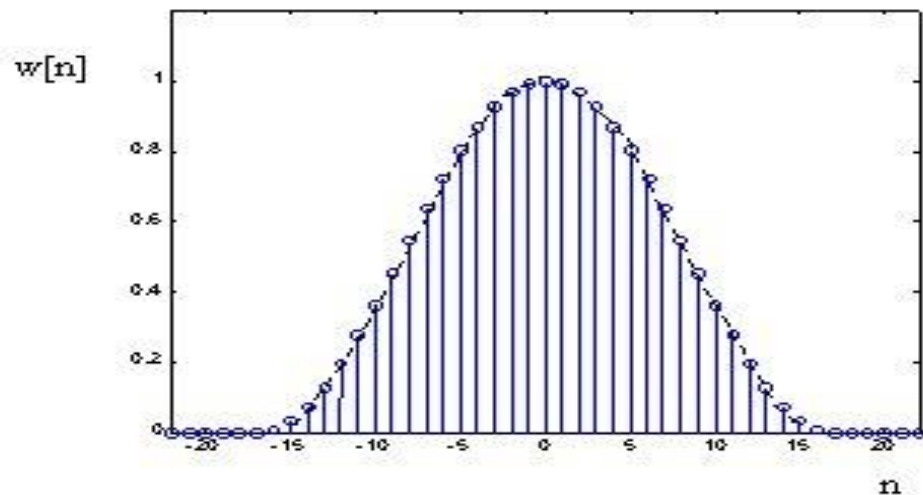
$$W_R(e^{j\omega}) = \frac{\sin(\omega(2M+1)/2)}{\sin(\omega/2)}$$



• Hanning window

$$w[n] = \frac{1}{2} \left[1 + \cos\left(\frac{\pi n}{M}\right) \right], -M \leq n \leq M$$

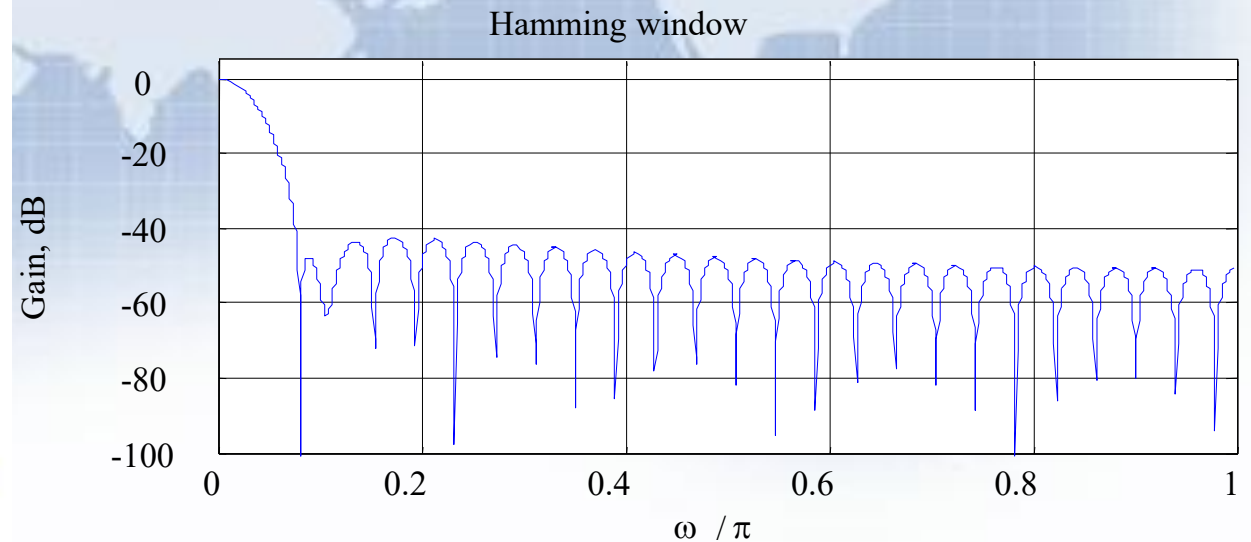
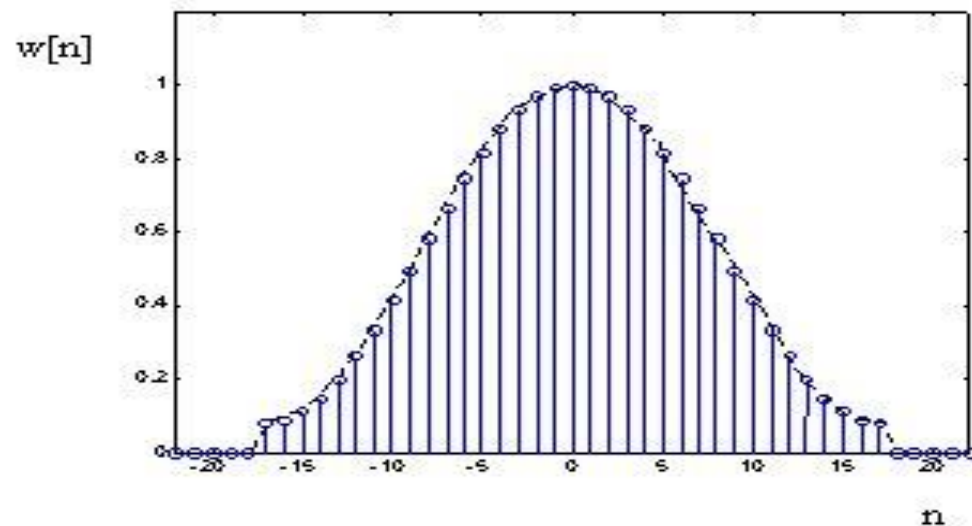
$$W(\omega) = \frac{1}{2} W_R(\omega) + \frac{1}{4} \left[W_R\left(\omega - \frac{\pi}{M}\right) + W_R\left(\omega + \frac{\pi}{M}\right) \right]$$



- Hamming window:

$$w[n] = \left[0.54 + 0.46 \cos\left(\frac{\pi n}{M}\right) \right], -M \leq n \leq M$$

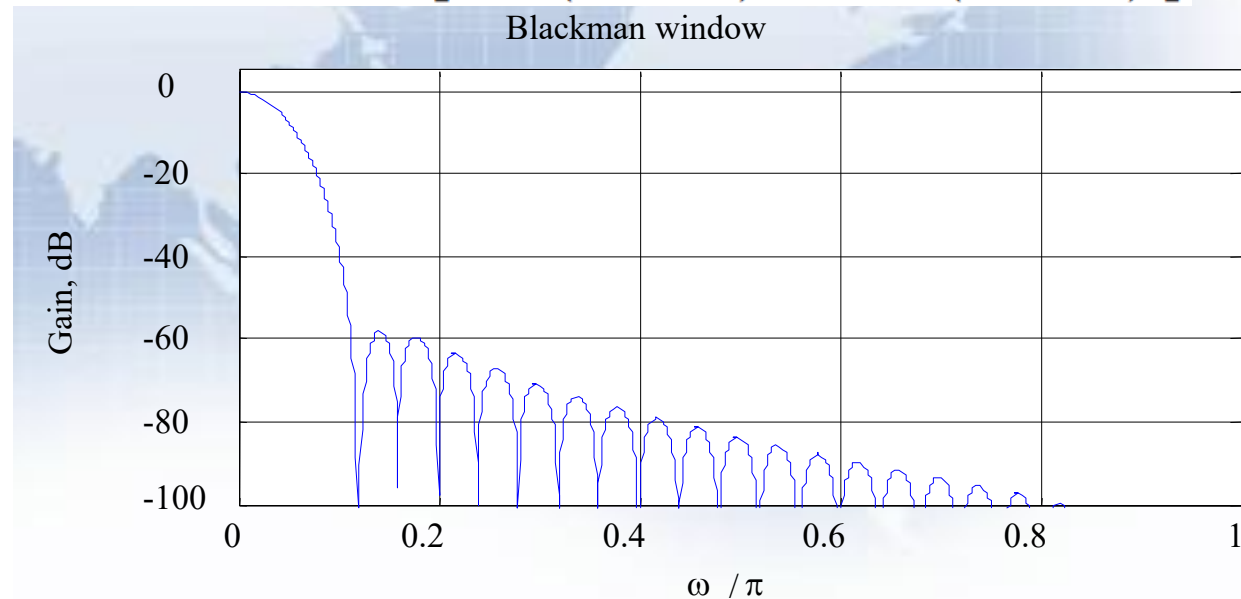
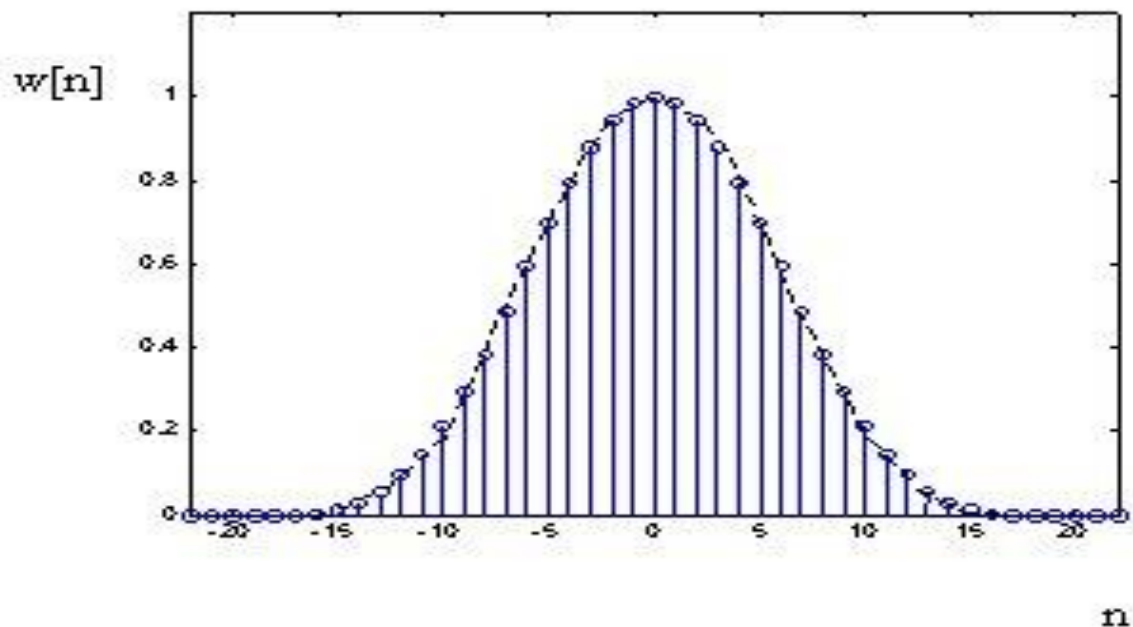
$$W(\omega) = 0.54W_R(\omega) + 0.23 \left[W_R\left(\omega - \frac{\pi}{M}\right) + W_R\left(\omega + \frac{\pi}{M}\right) \right]$$



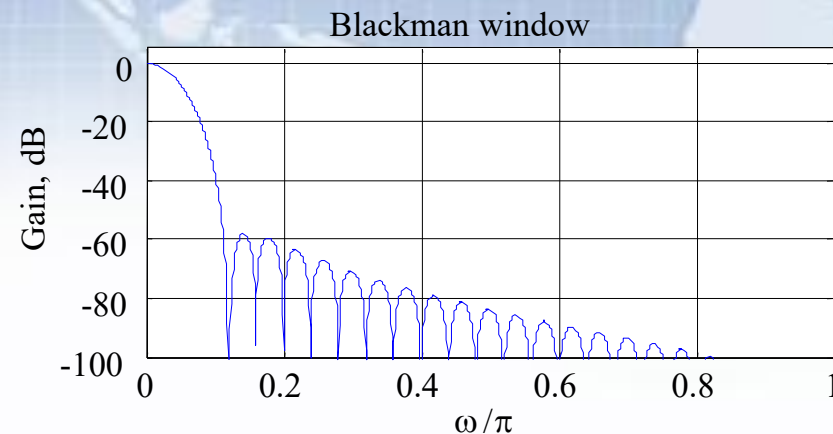
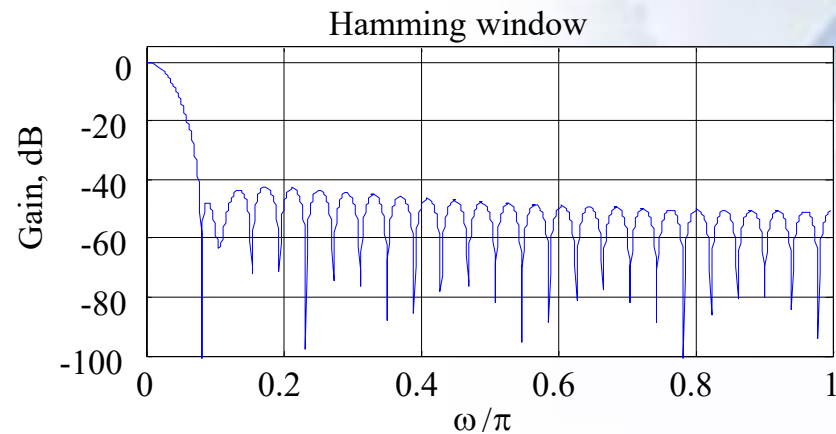
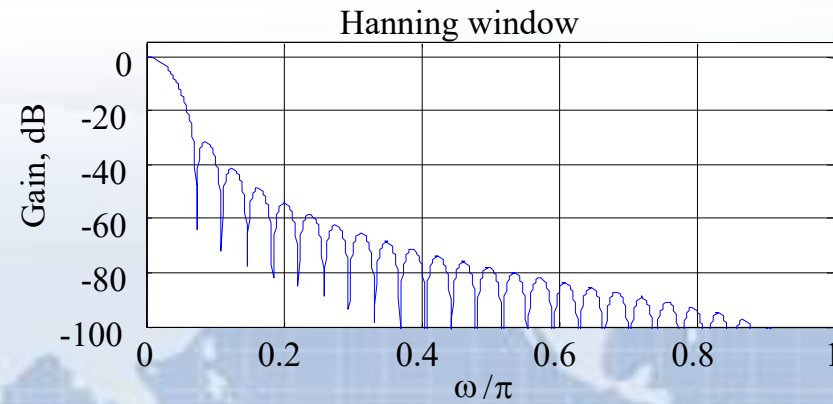
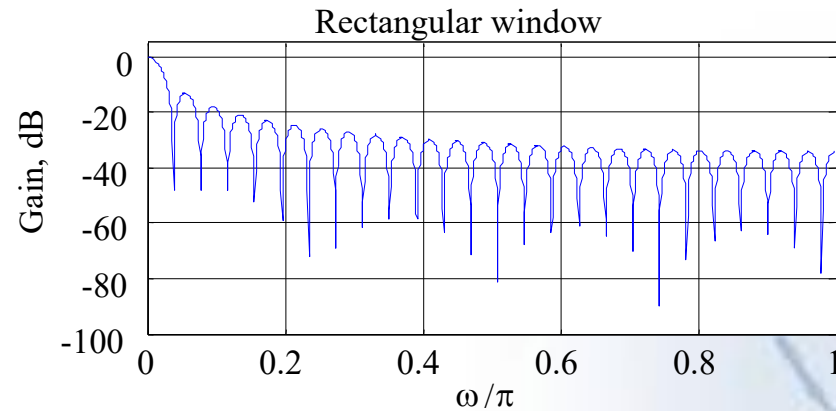
• Blackman window:

$$w[n] = \left[0.42 + 0.5 \cos\left(\frac{\pi n}{M}\right) + 0.08 \cos\left(\frac{2\pi n}{M}\right) \right] \quad -M \leq n \leq M$$

$$W(\omega) = 0.42W_R(\omega) + 0.25 \left[W_R\left(\omega - \frac{\pi}{M}\right) + W_R\left(\omega + \frac{\pi}{M}\right) \right] + 0.04 \left[W_R\left(\omega - \frac{2\pi}{M}\right) + W_R\left(\omega + \frac{2\pi}{M}\right) \right]$$



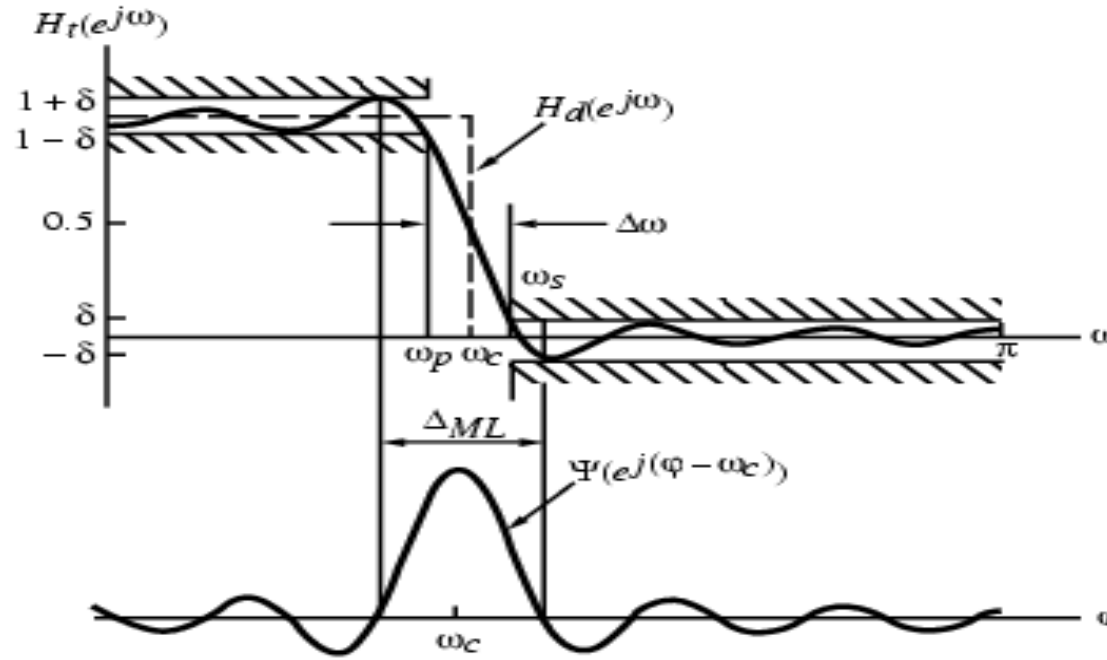
- Plots of magnitudes of the DTFTs of these windows for $M = 25$ are shown below:



Performance of a window function in LPF design



- Magnitude spectrum of each window characterized by a main lobe centered at $w = 0$ followed by a series of sidelobes with decreasing amplitudes.
- Parameters predicting the performance of a window in filter design are:
 - Main lobe width --- Δ_{ML}
 - Relative sidelobe level --- A_{sl}
 - given by the difference in dB between amplitudes of largest sidelobe and main lobe.



- **Observe** $\left| H_t(e^{j(\omega_c + \Delta\omega)}) + H_t(e^{j(\omega_c - \Delta\omega)}) \right| \cong 1$

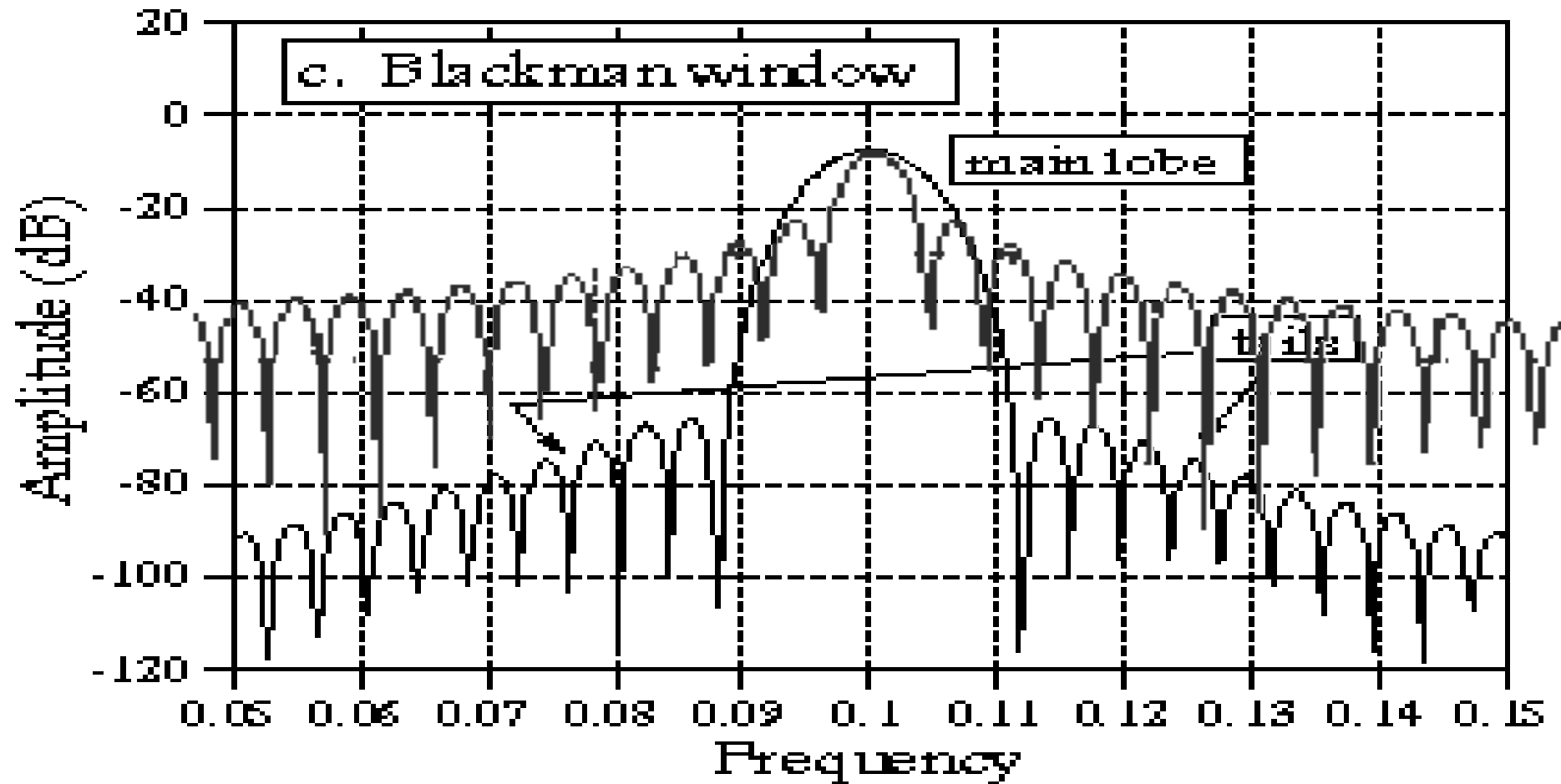
Thus, $|H_t(e^{j\omega_c})| \cong 0.5$

- **Passband and stopband ripples are the same.**



- Distance between the locations of the maximum passband deviation and minimum stopband value $\cong \Delta_{ML}$
- Width of transition band:
$$\Delta w = \omega_s - \omega_p < \Delta_{ML}$$
- To ensure a fast transition from passband to stopband, window should have a very small main lobe width.
- To reduce the passband and stopband ripple δ , the area under the sidelobes should be very small.
- Unfortunately, these two requirements are contradictory.

Rectangular and Blackman windows





- In the case of rectangular, Hann, Hamming, and Blackman windows, the value of ripple does not depend on filter length or cutoff frequency ω_c , and is essentially constant.
- In addition, the transition bandwidth is approximately given by

$$\Delta\omega \approx c / M$$

where c is a constant for most practical purposes.

→ So, we can determine $\Delta\omega$ by M .



Table 10.2 Properties of some fixed window functions

Type of windows	Main lobe width Δ_{ML}	Relative sidelobe level A_{sl}	Minimum stopband attenuation α_s	Transition bandwidth Δ_ω
Rectangular	$4\pi/(2M+1)$	13.3dB	20.9dB	$0.92\pi/M$
Hann	$8\pi/(2M+1)$	31.5dB	43.9dB	$3.11\pi/M$
Hamming	$8\pi/(2M+1)$	42.7dB	54.5dB	$3.32\pi/M$
Blackman	$12\pi/(2M+1)$	58.1dB	75.3dB	$5.56\pi/M$



通带上偏差: $A_{\max} = 20 \log_{10}(1 + \delta_p)$

通带下偏差: $A_{\min} = 20 \log_{10}(1 - \delta_p)$.

通带峰-峰ripple: $A_{pp} = 20 \log_{10} \frac{1 + \delta_p}{1 - \delta_p}$

Window	α_s	δ_p	通带上偏差	通带下偏差	峰峰值 ripple
Rectangular	20.9 dB	0.0902	+0.750 dB	−0.821 dB	1.570 dB
Hann	43.9 dB	0.00638	+0.0553 dB	−0.0556 dB	0.1109 dB
Hamming	54.5 dB	0.00188	+0.0163 dB	−0.0164 dB	0.0327 dB
Blackman	75.3 dB	0.000172	+0.00149 dB	−0.00149 dB	0.00298 dB

10.2.5 Adjustable Window Functions



- Rectangular, Hann and Hamming **have constant ripple**, independent of filter length or cut off frequency ω_c

- Kaiser window

$$w[n] = \frac{I_0(\beta \sqrt{1 - [n/M]^2})}{I_0(\beta)}, \quad -M \leq n \leq M$$

where $I_0(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{jx \sin y} dy \approx 1 + \sum_{k=1}^{\infty} \left[\frac{(x/2)^k}{k!} \right]^2$

- β controls the minimum attenuation α_s in the stopband
- So, it is called as **adjustable window function**.

10.2.5 Adjustable Window Functions



- **Kaiser window**

- **The relationship between β and α_s :**

$$\beta = \begin{cases} 0.1102(\alpha_s - 8.7), & \text{for } \alpha_s > 50 \\ 0.5842(\alpha_s - 21)^{0.4} + 0.07886(\alpha_s - 21) & \text{for } 21 \leq \alpha_s \leq 50, \\ 0, & \text{for } \alpha_s < 21 \end{cases}$$

- **The filter order N :**

$$N = \frac{\alpha_s - 8}{2.285(\Delta\omega)}$$

- **Dolph-Chebyshev window**

$$w[n] = \frac{1}{2M+1} \left[\frac{1}{\gamma} + 2 \sum_{k=1}^M T_{2M} \left(\beta \cos \frac{k\pi}{2M+1} \right) \cos \frac{2nk\pi}{2M+1} \right] \quad -M \leq n \leq M$$

where

$$\gamma = \frac{\text{sidelobe amplitude}}{\text{main lobe amplitude}}$$

$$\beta = \cosh \left(\frac{1}{2M} \cosh^{-1} \frac{1}{\gamma} \right)$$

$$T_l(x) = \begin{cases} \cos(l \cos^{-1} x), & \text{for } |x| \leq 1 \\ \cosh(l \cosh^{-1} x), & \text{for } |x| > 1 \end{cases}$$

$$N = 2M = \frac{2.056\alpha_s - 16.4}{2.285(\Delta\omega)}$$

Example of FIR Filter Design:



Design a linear phase FIR LPF:

- If given the passband edge frequency $\omega_p=0.3\pi$, the passband ripple is 1dB, the stopband edge frequency $\omega_s=0.5\pi$, the minimum stopband attenuation $<-40\text{dB}$.

• **Solution:**

$$\alpha_p = 1\text{dB}, \alpha_s = 40\text{dB}$$

$$\omega_c = (\omega_p + \omega_s) / 2 = 0.4\pi$$



The ideal LP impulse response is:

$$h_{LP}[n] = \frac{\sin \omega_c n}{\pi n}, \quad -\infty < n < \infty$$

Select a window based on table 10.2 and specifications: $\alpha_s=40\text{dB}$;

If we select window Hann, calculate order N:

$$\Delta\omega = \omega_s - \omega_p = 0.2\pi = \frac{3.11\pi}{M} \Rightarrow M = 16, N = 2M = 32$$

$$h[n] = \frac{\sin[0.4\pi(n-16)]}{\pi(n-16)} w_{Hann}[n-16], \quad 0 \leq n \leq 32$$



If we select window Hamming:

$$\Delta\omega = 0.2\pi = \frac{3.32\pi}{M} \Rightarrow M = 17, N = 2M = 34$$

If we select window Blackman:

$$\Delta\omega = 0.2\pi = \frac{5.56\pi}{M} \Rightarrow M = 28, N = 2M = 56$$

If we select window Kaiser:

$$N = \frac{\alpha_s - 8}{2.285(\Delta\omega)} \cong 23,$$

$$\beta = 0.5842(\alpha_s - 21)^{0.4} + 0.07886(\alpha_s - 21) = 3.3953$$

FIR Filter Design Example 1



- **The specifications of a low pass filter:**

Pass band edge 400Hz

Stop band edge 1650Hz

Stop band attenuation 50dB

Sampling frequency 11.025kHz



- **Transition width $\Delta f = 1650 - 400 = 1250\text{Hz}$**

$$\Delta\omega = 2\pi \times \Delta f / F_T = \frac{2\pi \times 1250}{11025} = 0.22676\pi = 0.71238$$

- **Another parameter $f_c = (400 + 1650)/2 = 1025\text{ Hz}$, So**

$$\omega_c = 2\pi \times f_c / F_T = \frac{2\pi \times 1025}{11025} = 0.185941\pi = 0.584151$$

- **Choose Hamming window (-54dB),**



$$M \approx \frac{c}{\Delta\omega} = \frac{3.32\pi}{0.71238} = 14.6 \approx 15 \quad \therefore N=2M=30$$

- **The finite-length causal FIR LP filter:**

$$\hat{h}_{LP}[n] = \frac{\sin(\omega_c(n-M))}{\pi(n-M)}, 0 \leq n \leq N$$



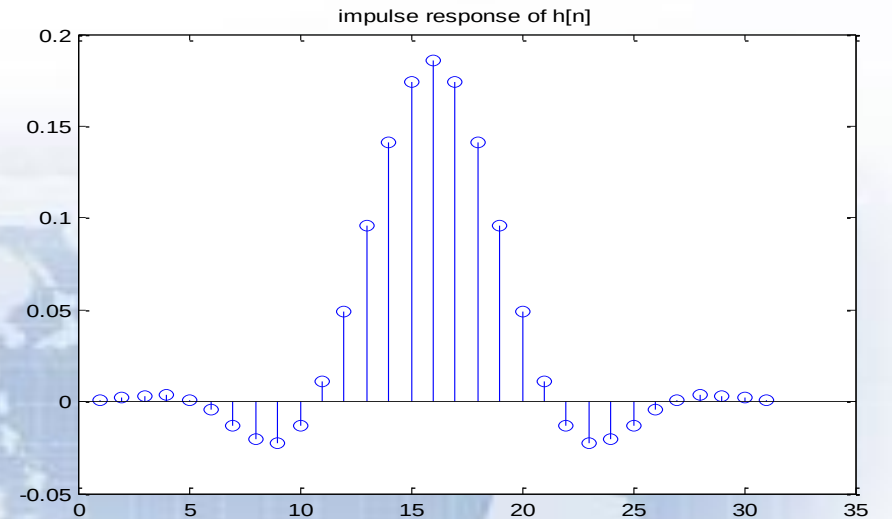
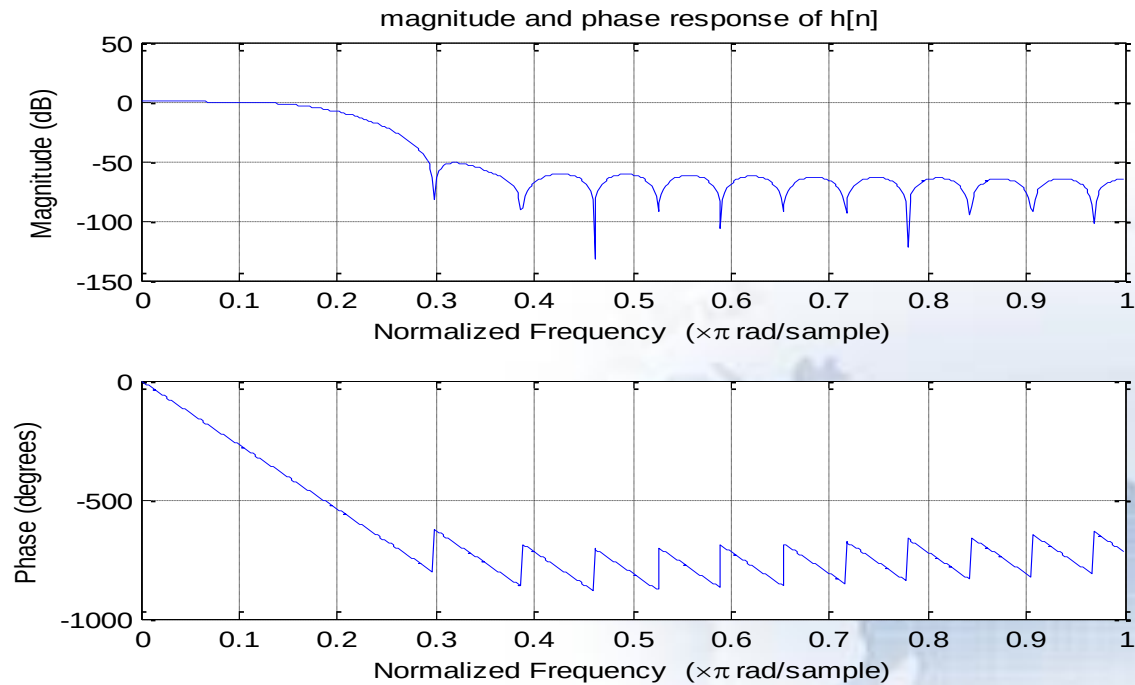
- **The Hamming window is**

$$\omega[n] = 0.54 - 0.46 \cos\left(\frac{\pi(n-M)}{M}\right), 0 \leq n \leq N$$

- **So, $h[n]$ of the filter which we need is**

$$h[n] = \hat{h}_{LP}[n] \cdot \omega[n], 0 \leq n \leq N-1$$

- **Then the magnitude response and the impulse response are drawn in next slide:**



So, the original sound: 
the output sound after passing the LP filter: 

FIR Filter Design Example 2



- **The specifications of a high pass filter:**

Pass band edge 1650Hz

Stop band edge 400Hz

Stop band attenuation 50dB

Sampling frequency 11.025kHz



- **Transition width $\Delta f = 1650 - 400 = 1250 \text{ Hz}$**

$$\Delta\omega = 2\pi \times \Delta f / F_T = \frac{2\pi \times 1250}{11025} = 0.22676\pi = 0.71238$$

- **Another parameter $f_c = (400 + 1650)/2 = 1025 \text{ Hz}$,**

$$\omega_c = 2\pi \times f_c / F_T = \frac{2\pi \times 1025}{11025} = 0.185941\pi = 0.584151$$

- **Choose Hamming window (-54dB),**

$$M \approx \frac{c}{\Delta\omega} = \frac{3.32\pi}{0.71238} = 14.6 \approx 15 \quad \therefore N=2M=30$$

- **The finite-length causal FIR HP filter:**

$$\hat{h}_{HP}[n] = \begin{cases} -\frac{\sin(\omega_c(n-M))}{\pi(n-M)}, & 0 \leq n \leq N, \quad n \neq M \\ 1 - \frac{\omega_c}{\pi}, & n = M \end{cases}$$



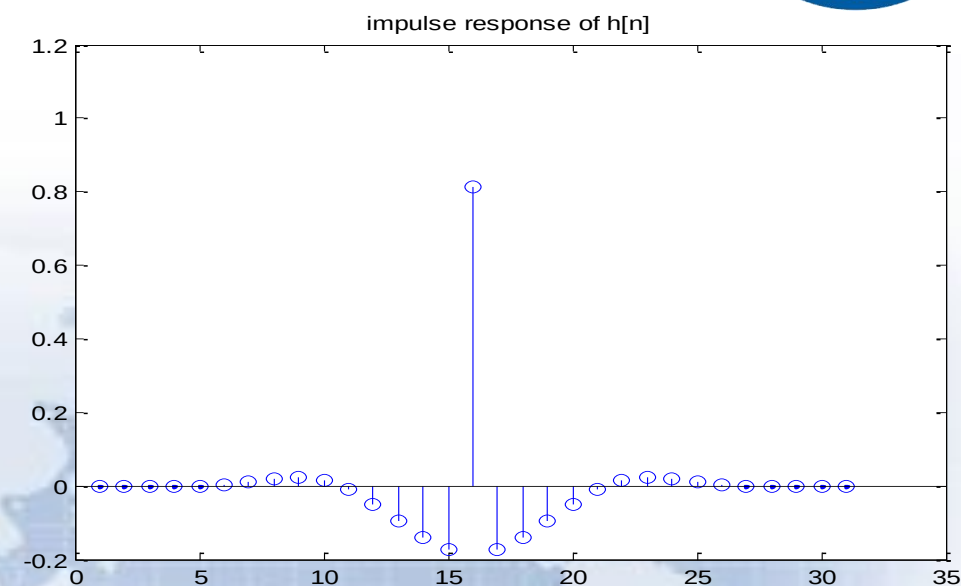
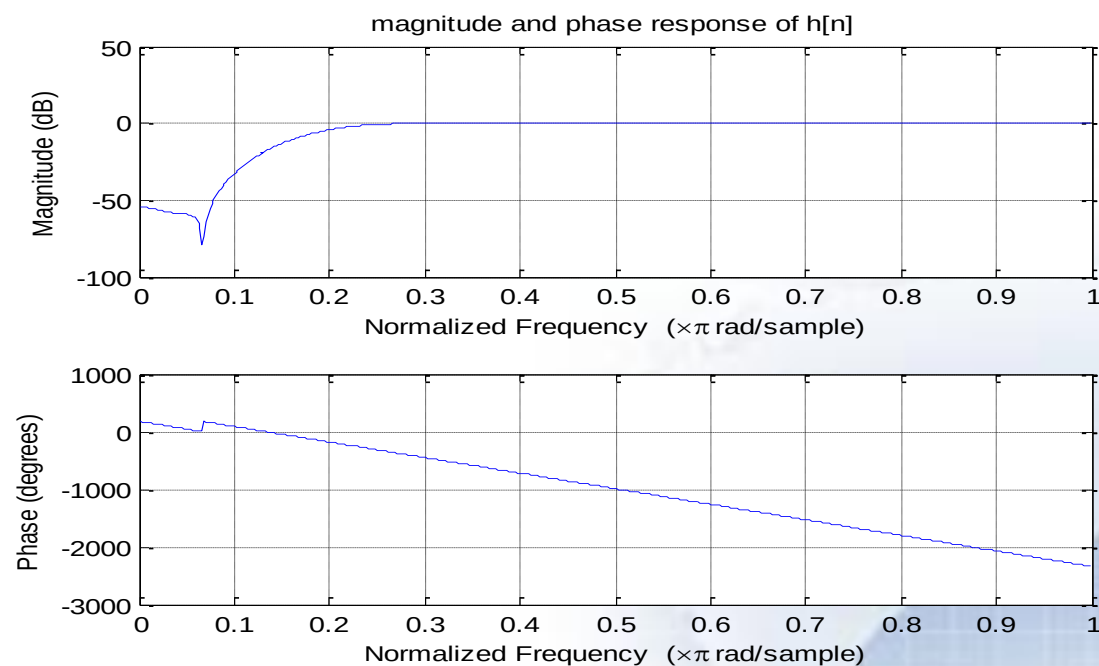
- The Hamming window is (no change)

$$\omega[n] = 0.54 + 0.46 \cos\left(\frac{\pi(n-M)}{M}\right), 0 \leq n \leq N$$

- So, $h[n]$ of the filter which we need is

$$h[n] = \hat{h}_{LP}[n] \cdot \omega[n], 0 \leq n \leq N-1$$

- Then the magnitude response and the impulse response are drawn in next slide:





- Find an approximation of the desired FR $H(e^{j\omega})$ in frequency domain.

➤ The method:

Let $H(k) = H(e^{j\frac{2\pi}{N+1}k})$ denote the sampling values of N+1-point FIR filter $H(e^{j\omega})$

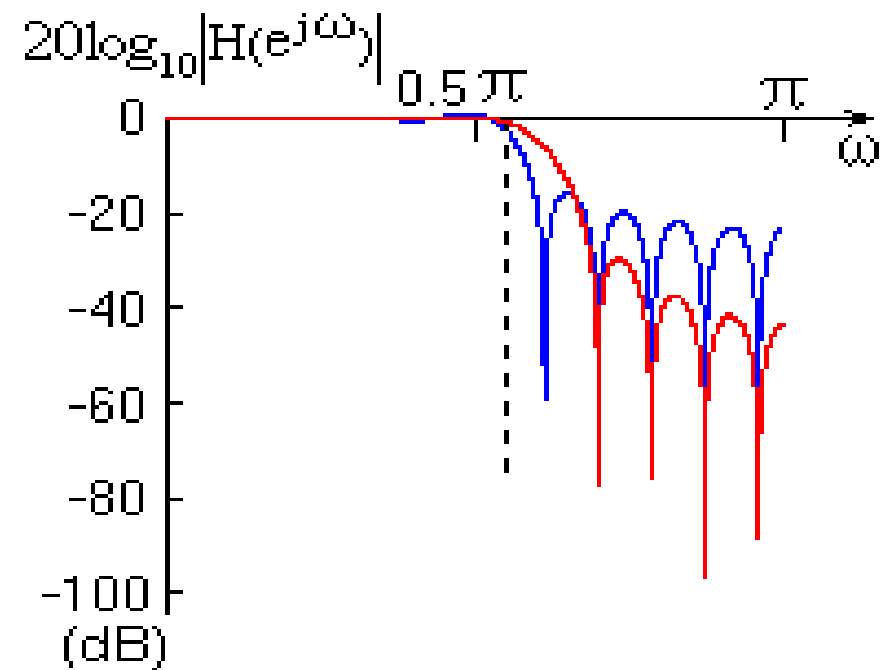
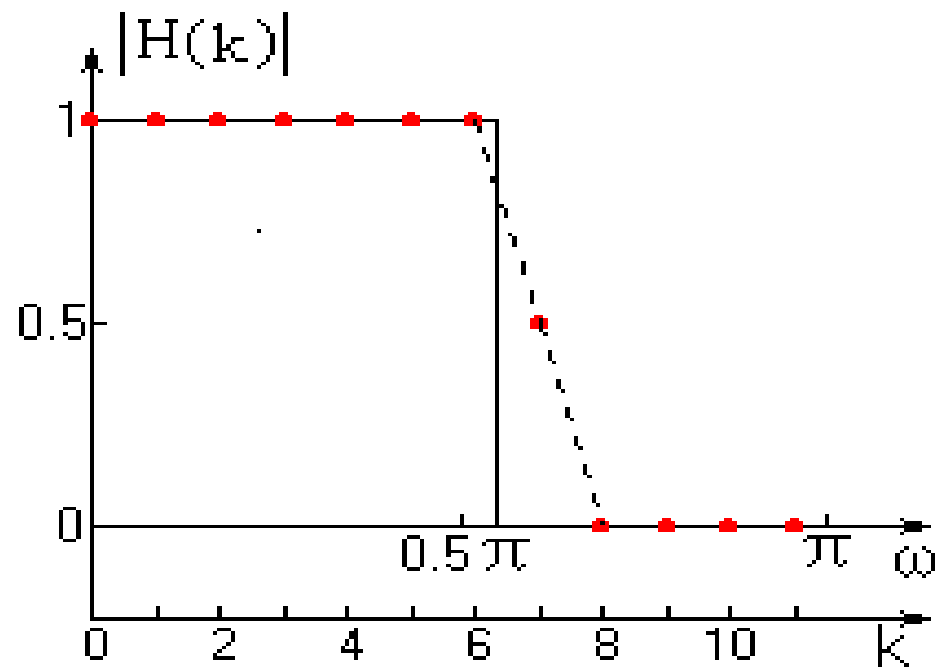
$$H(k) \xrightarrow{\text{approximation}} H_d(k) = H_d(e^{j\frac{2\pi}{N+1}k})$$

$$H(e^{j\omega}) \xrightarrow{\text{approximation}} H_d(e^{j\omega})$$

Use $H[k] = H_d(e^{j\omega})|_{\omega=\omega_k=2\pi/M}$ to approximate the desired frequency response $H_d(e^{j\omega})$, then from: $h[n] = \text{IDFT}[H[k]]$ we can get the length-N impulse response $h[n]$ of the FIR filter.

- Note: $H[k]$ must satisfy the constraint that the FIR filter is linear phase.**

For example, a LPF with one sample of magnitude on the transition band is shown below.



10.5 FIR Digital Filter Design Using MATLAB



- The *Signal Processing Toolbox* of MATLAB includes a variety of M-files for the design of FIR digital filters
- We will illustrate the use of some these functions in the following.
- Also, we can use the GUI design tools in MATLAB **FDATool** command !



- **The window-based FIR filter design process involves three steps:**
 1. **The type of windows to be used is determined to meet the desired stop-band attenuation or pass-band ripple.**
 2. **The order of the FIR is estimated by using Table 10.2 or Eq.(10.45) for Kaiser window.**
 3. **The desired impulse response of the ideal filter is computed, which is then multiplied by the window coefficients.**



- The M-file **kaior**d determines the FIR filter order using Kaiser's formula.
- The M-file **bellangord** determines the FIR filter order using Bellanger's formula.
- The M-file **remezord** determines the FIR filter order using Hermann's formula.
- Using the Kaiser window, the order can be estimated using the formula of (10.45). The M-file **kaiserord** can be employed.



- The functions available in MATLAB for the design of FIR filters using the win-dows Fourier approach are **fir1** and **fir2**.
- The function **fir1** is used to design conventional lowpass, highpass, bandpass, bandstop and multiband FIR.
- The function **fir2** is employed to design FIR with arbitrarily shaped magnitude response.



Window Generation

`W=hann(L)`

`W=hamming(L)`

`W=blackman(L)`

`W= Kaiser(L,beta)`

Filter Design

`b=fir1(N,Wn) %N-order, Wn=[w1,w2,...]`

`b=fir1(N,Wn,'ftype') %ftype—lpf,hpf,..`

`b=fir1(N,Wn>window) %default hamming`



Example 10.25

```
% Lowpass Filter Design Using the Kaiser Window
fpts = input('Type in the bandedges = ');
mag = input('Type in the desired magnitude values = ');
dev = input('Type in the ripples in each band = ');
[N, Wn, beta, ftype] = kaiserord(fpts,mag,dev)
kw = kaiser(N+1,beta);
b = fir1(N,Wn, kw);
[h,omega] = freqz(b,1,512);
plot(omega/pi,20*log10(abs(h)));grid;
xlabel('\omega/\pi'); ylabel('Gain, dB');
```



Type in the bandedges = [0.3 0.4]

Type in the desired magnitude values = [1 0]

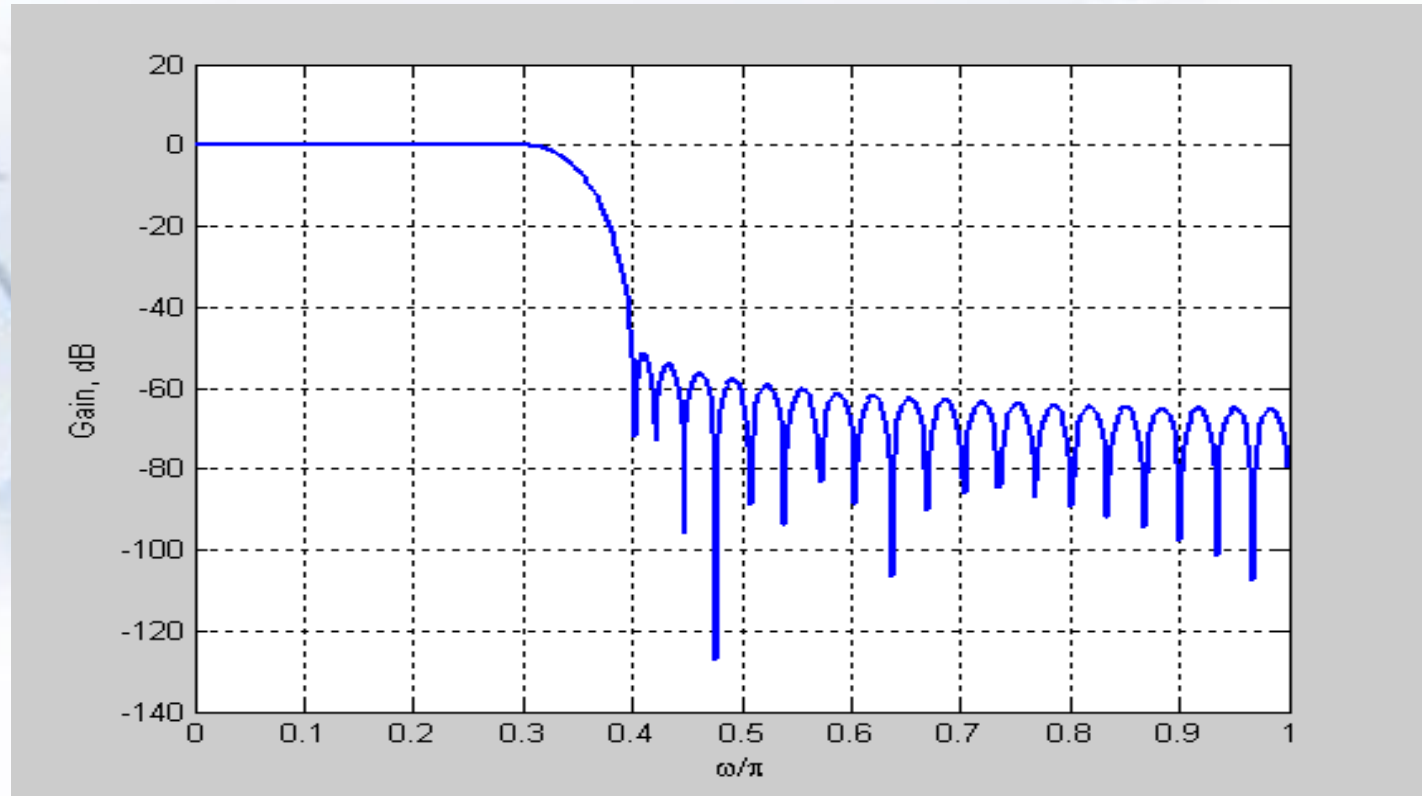
Type in the ripples in each band = [0.003162 0.003162]

N = 59

Wn = 0.3500

beta = 4.5513

ftype = low





Example 10.26

Use the same program as Example 10.25
by modifying `b = fir1(N,Wn,'high',kw);`

Type in the bandedges = [0.4 0.55]

Type in the desired magnitude values = [0 1]

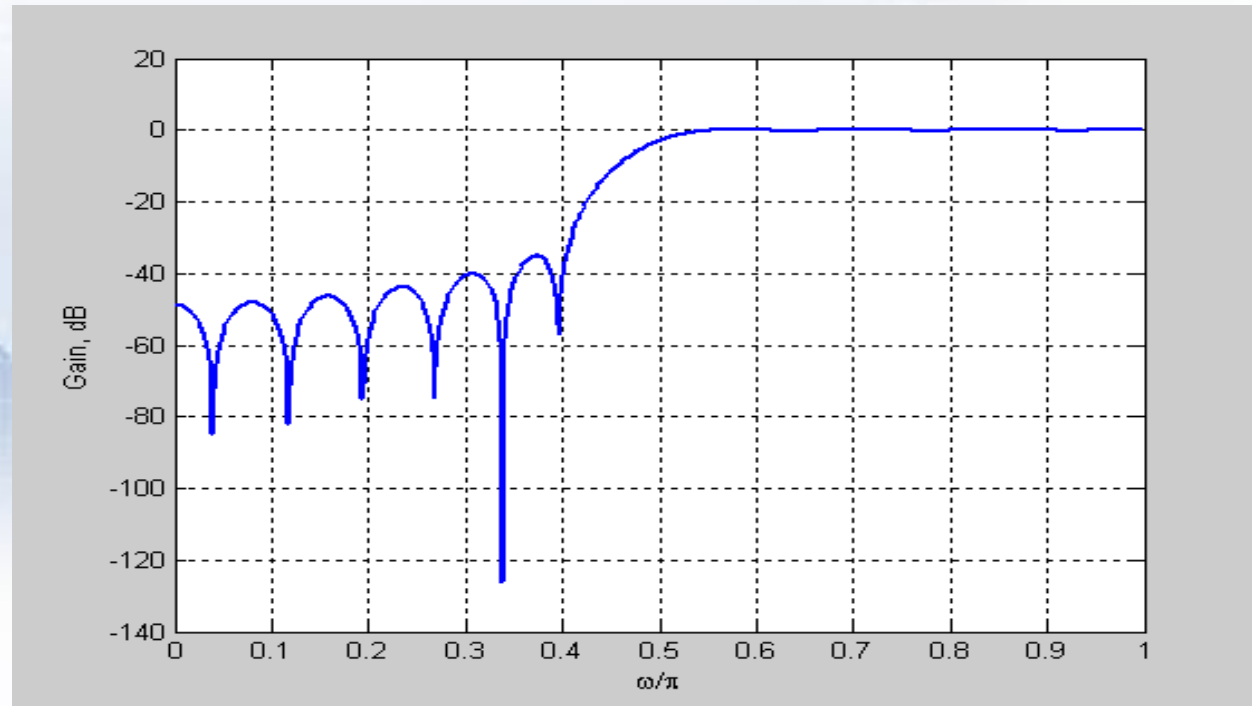
Type in the ripples in each band = [0.02 0.02]

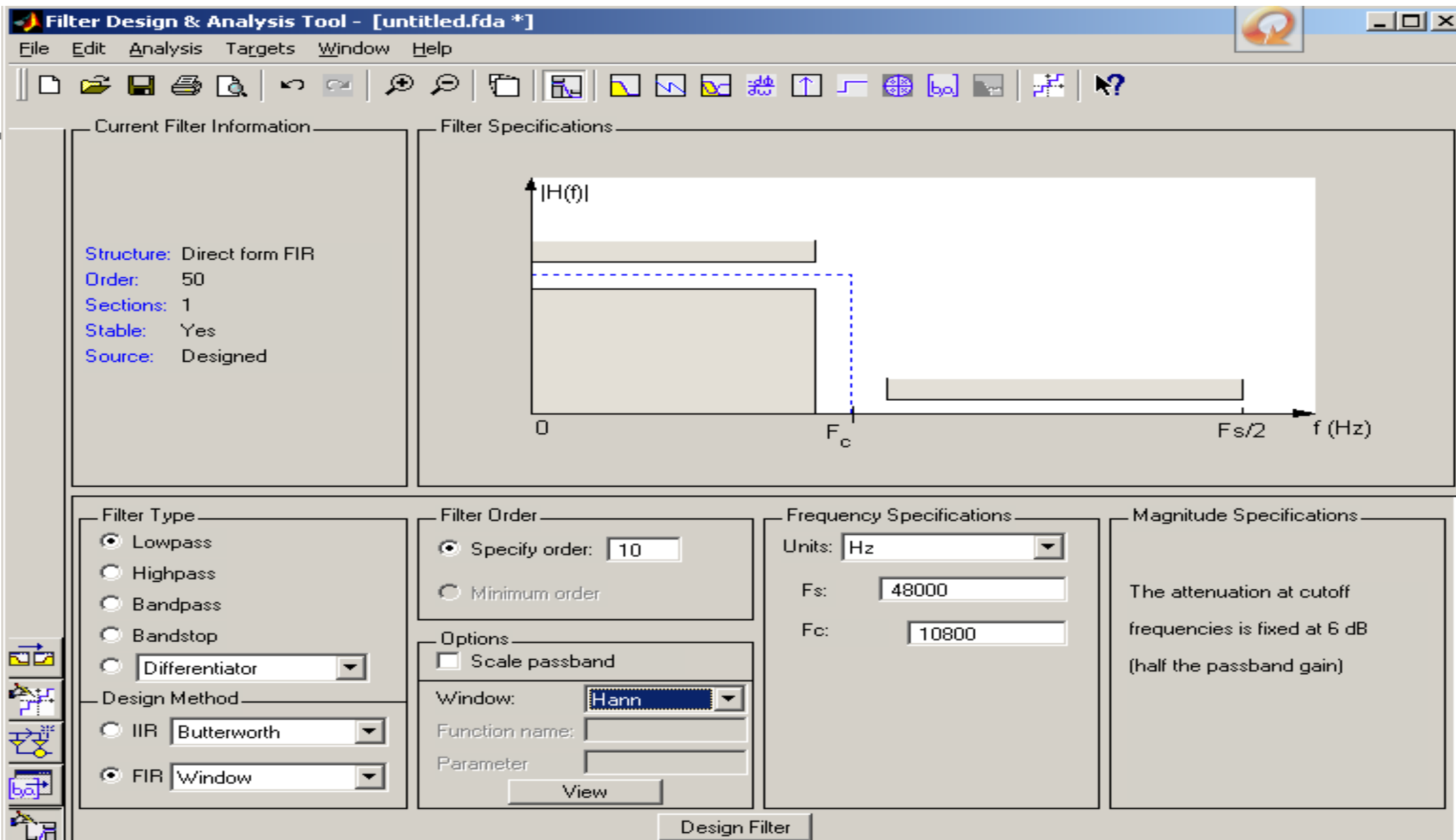
`N = 26`

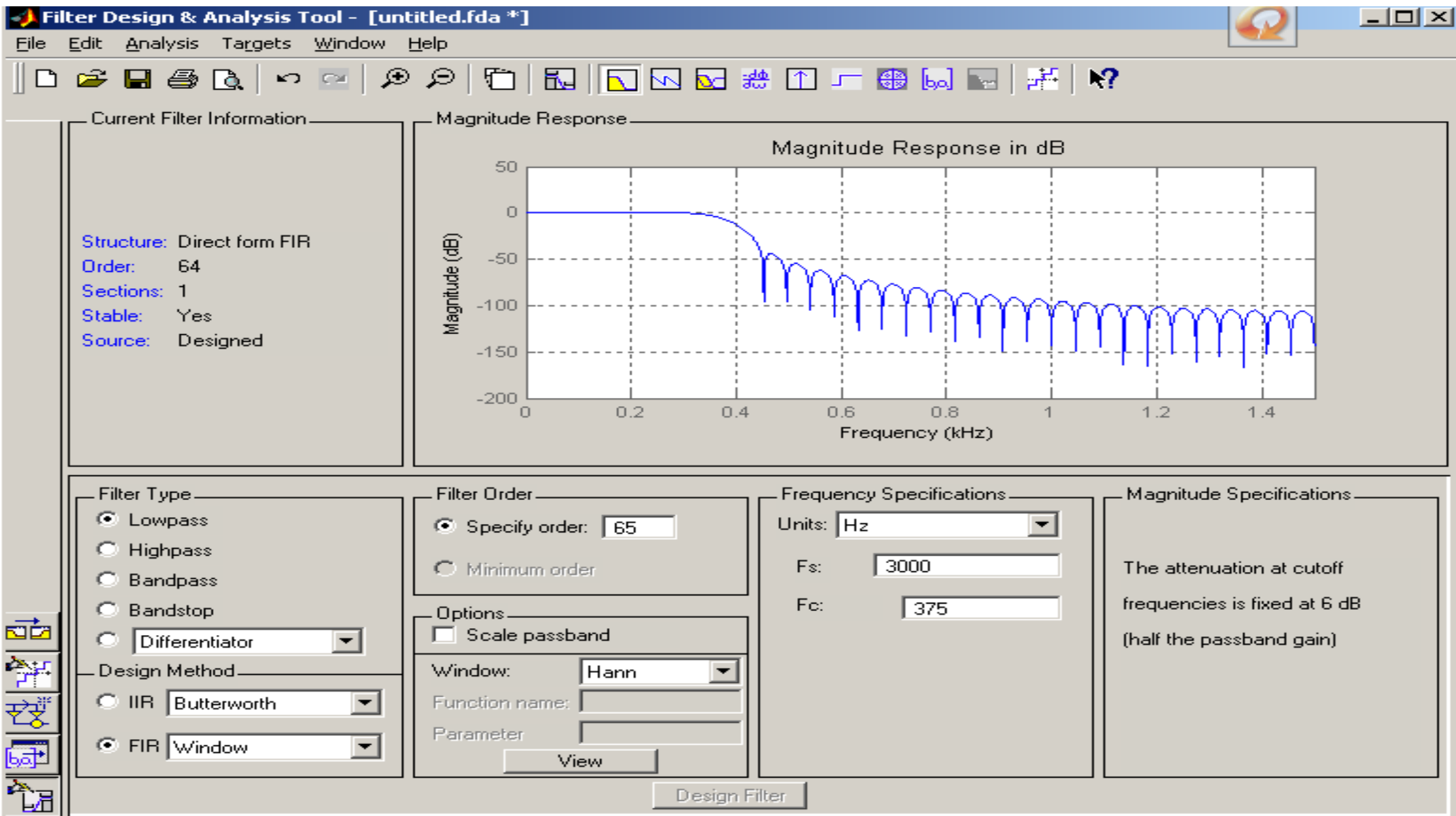
`Wn = 0.4750`

`beta = 2.6523`

`ftype = high`







Homework:



- 10.1, 10.17, 10.30, 10.31
M10.8, M10.9, M10.17